

# FAKE NEWS DETECTION USING MACHINE LEARNING AND NLP

## Submitted By

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## Abstract

Fake news has become a major issue in digital media and social networking platforms. This project aims to develop a Fake News Detection system using Machine Learning and Natural Language Processing techniques. The system uses TF-IDF Vectorization and Logistic Regression to classify news articles as REAL or FAKE.

## Dataset Preview

| text                                                        | label |
|-------------------------------------------------------------|-------|
| Government announces new education policy for schools       | REAL  |
| Scientists discover water on Mars after successful research | REAL  |
| Indian cricket team wins the international tournament       | REAL  |
| New railway line inaugurated in Tamil Nadu                  | REAL  |
| Stock market rises after positive economic reports          | REAL  |

**Dataset Shape:** (30, 2)

### Label Distribution:

REAL : 15  
FAKE : 15

## Methodology

1. Data Collection
2. Data Preprocessing
3. Feature Extraction using TF-IDF
4. Model Building using Logistic Regression
5. Training and Testing
6. Prediction and Evaluation

## Model Evaluation

**Accuracy:** 0.5000

| Class | Precision | Recall | F1-Score | Support |
|-------|-----------|--------|----------|---------|
|-------|-----------|--------|----------|---------|

|      |      |      |      |   |
|------|------|------|------|---|
| FAKE | 0.50 | 0.67 | 0.57 | 3 |
| REAL | 0.50 | 0.33 | 0.40 | 3 |

## Confusion Matrix

|             | Predicted FAKE | Predicted REAL |
|-------------|----------------|----------------|
| Actual FAKE | 2              | 1              |
| Actual REAL | 2              | 1              |

## Custom News Prediction

Input News:

Government announces new education policy for schools

Prediction Result:

REAL

## Sample Python Code

```
vectorizer = TfidfVectorizer(stop_words='english')
model = LogisticRegression()
model.fit(X_train_vectorized, y_train)
prediction = model.predict(news_vector)
```

## Conclusion

This project successfully developed a Fake News Detection system using Machine Learning and NLP techniques. The model was able to classify news articles as REAL or FAKE using TF-IDF Vectorization and Logistic Regression.